

“CHARGE AND DISCHARGE OF CAPACITOR”

- Student Name:
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- Date of Experiment:.
- Date of Submission:

➤ **Abstract:**

The purpose of this experiment is to investigate the charging and the discharging of a capacitor. In this experiment a capacitor is charged and discharged and the time taken is recorded at equal intervals.

➤ **Objective:**

To investigate the charge and the discharge of a capacitor.

➤ **Introduction:**

A capacitor is a passive two-terminal electrical component used to store energy electrostatically in an electric field. The forms of practical capacitors vary widely but all contain at least two electrical conductors or plates separated by a dielectric or insulator. The conductors can be thin film, foil etc. The insulator acts to increase the capacitor's charge capacity. An insulator can be glass, ceramic, plastic film, air, paper etc.

When a capacitor is connected to a circuit with a DC supply two processes occur which are called as charging and discharging.

Charging: When a capacitor is connected to the DC supply and current starts to flow through the circuit both plates of the capacitor gets the equal and opposite charges and an increasing potential difference which is created while the capacitor is charging. Once the voltage at the terminals of the capacitor is equal to the power supply the capacitor is fully charged and the current stops flowing through the circuit, the charging phase is over.

Discharging: When the capacitor is disconnected from the power supply the capacitors starts to discharge through the resistor and the voltage between the plates drops gradually to zero.

Resistor: A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. Resistors act to reduce current flow, and, at the same time, act to lower voltage levels within circuits.

➤ **Apparatus:**

- Power supply
- Capacitor
- Resistor

- Voltmeter
- Stop Watch

➤ **Procedure:**

- Construct the circuit as shown below in Figure 1.

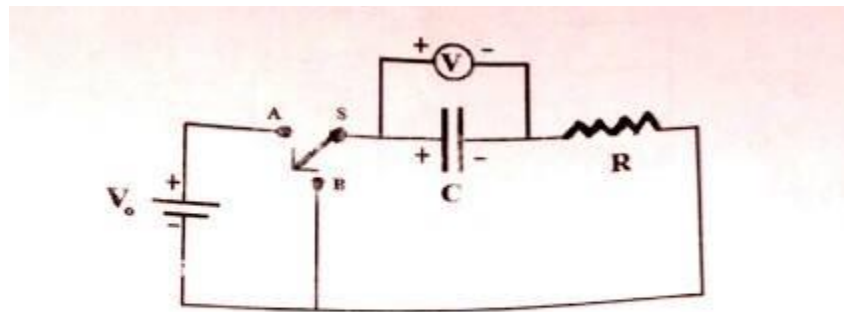


Figure 1

- **Case 1:** When the switch S is placed in position A , the charge flows from the battery into the capacitor until the capacitor is fully charged.
- **Case 2:** When the switch is placed in B , the capacitor discharges through the resistor.
- Record values of voltages V and current I and corresponding time t in each case

➤ **Results:**

Table 1

Charging of a capacitor		
Current (I)	Voltage (V)	Time (t)
		0
		5
		10
		15
		20
		25
		30
		35
		40
		45

Table 2

Discharging of a capacitor		
Current (I)	Voltage (V)	Time (t)
		0
		5

		10
		15
		20
		25
		30
		35
		40
		45

In this experiment, the current flowing through a resistor will be measured as the voltage across the resistor is varied.

So please fill the Table 1 for this circuit.

6) At time $t=0$ when we first close the switch S in the circuit, the capacitor has no charge, and so the current I will be determined by the resistor alone. The capacitor here acts as a short circuit. At any later time, the charge will start to increase while the current decrease.

Then, $q(t)$ will reach the constant value of $q = VC$. At this instant the capacitor will be fully charged.

The current, on the other hand, will be zero at this instant. At this step try to fill the Table 1.

7) Find the experimental time constant τ of the circuit from the I vs t graphs. Find it from both the charging and discharging graphs. Then, compare experimental time constant τ

with its theoretical value obtained by

$$t = RC$$

➤ **Conclusion:**